Beyond the average phenotype: individual variation alters mechanistic forecasts of climate responses

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Running Title: Individual variation alters forecasts

Abstract

Mechanistic models are central to forecasting biological responses to climate change because they simulate organism–environment interactions that determine growth, survival and reproduction. Yet these models generally represent species by a single average phenotype, despite substantial physiological variation among individuals. Because organismal performance emerges from nonlinear interactions, Jensen's inequality predicts that forecasts for an average phenotype may differ systematically from those obtained by integrating across heterogeneous individuals. Whether the magnitude of this bias changes across environments and warming scenarios remains largely unknown. Here, we combined longitudinal measurements of growth and metabolism from individually monitored alpine newts (*Ichthyosaura alpestris*) with Dynamic Energy Budget (DEB) models to estimate physiological parameters for an average phenotype and for individual phenotypes. We coupled these parameterisations with biophysical models to predict twelve life-history and physiological performance traits across Europe under current climate and +2 °C and +4 °C warming scenarios. We then compared average-phenotype forecasts with forecasts integrating across individuals to quantify the bias arising from nonlinear physiological averaging. Average-phenotype forecasts consistently diverged from heterogeneous-population forecasts, but the magnitude and sign of this bias depended on the performance trait, geographic location and warming scenario. Reproductive traits were more sensitive to physiological averaging than structural traits, while warming amplified, attenuated or reversed bias depending on trait and environmental context. Nevertheless, both parameterisations recovered the same broad spatial pattern of biological responses to climate warming, but differed in the magnitude of the predicted changes. Our results identify biological heterogeneity as an important, context-dependent source of uncertainty in mechanistic climate forecasting. Representing populations as distributions of physiological phenotypes rather than single average individuals provides a means to propagate this heterogeneity through forecasts and to identify the traits, regions and climate scenarios for which predictions are most sensitive to nonlinear averaging.

Keywords: biophysical modelling; climate change; Dynamic Energy Budget theory; intraspecific variation; Jensen's inequality; niche modelling

1 Introduction

Predicting how populations will respond to climate change remains one of the central challenges in ecology. Increasingly, these predictions rely on mechanistic approaches that simulate how organisms interact with their abiotic environment, thereby linking environmental conditions to organismal performance (Briscoe et al., 2023; Kearney & Porter, 2009; Kearney & Porter, 2017, 2020). By coupling organismal functional traits (e.g., morphology, physiology and behaviour) with microclimatic conditions derived from first principles, mechanistic models generate biologically grounded forecasts, overcoming some of the limitations of purely correlative approaches (Briscoe et al., 2023; Kearney & Porter, 2009; Urban et al., 2016), especially when environmental conditions extend beyond those currently experienced by species (Elith & Leathwick, 2009; Maino et al., 2016). Mechanistic forecasts have therefore become increasingly realistic in their representation of environmental heterogeneity. Yet, organismal heterogeneity is rarely treated with the same resolution (but see Enriquez-Urzelai et al., 2020; Kolbe et al., 2010; Riddell et al., 2022). How well this simplification captures population responses to environmental change remains largely unknown.

Individuals within populations frequently differ in morphology, physiology, behaviour and life-history traits, and these differences influence resource use, species interactions, demographic rates and ecosystem functioning (Bolnick et al., 2003, 2011; Des Roches et al., 2018). Among the many dimensions of individual variation, physiology is particularly relevant for understanding responses to environmental change because it provides the mechanistic link between environmental conditions and ecological performance (Seebacher & Franklin, 2012). Individuals commonly differ in metabolic rates, thermal sensitivities, or energy allocation and the plasticity of these traits (Gvoždík, 2024; Pettersen & Metcalfe, 2024). Consequently, environmental change acts on distributions of physiological phenotypes, whereas mechanistic forecasts commonly represent populations using a single average phenotype. Whether ignoring this variation biases forecasts, however, depends on how physiological variation interacts with environmental conditions to determine organismal performance.

The ecological consequences of consistent among-individual variation in physiological traits, however, cannot be understood from trait variation alone. Performance emerges from interactions between organismal phenotypes and environmental conditions. These interactions can be represented as multidimensional performance surfaces describing how a given performance trait varies across combinations of physiological phenotypes and abiotic conditions (Laughlin and Messier, 2015; Li et al., 2021; **Fig. 1a**). Such surfaces are rarely linear. Instead, they commonly exhibit thermal optima, thresholds, asymmetries and other nonlinear features that are pervasive in ecological systems (Bolnick et al., 2011; Ruel & Ayres, 1999). Consequently, the ecological consequences of individual variation depend not only on the amount of physiological variation within a population, but also on the environmental context in which it is expressed: the same distribution of physiological phenotypes may produce similar performance in some environments but markedly different performance in others (Ross et al., 2023). Moreover, different components of performance are likely to be associated with different performance surfaces. Growth, reproduction or survival might not share the same thermal optima, thresholds or nonlinearities (Kontopoulos et al., 2024; Sinclair et al., 2016), suggesting that the effects of physiological variation, and the biases introduced by physiological averaging, may differ among components of organismal performance.

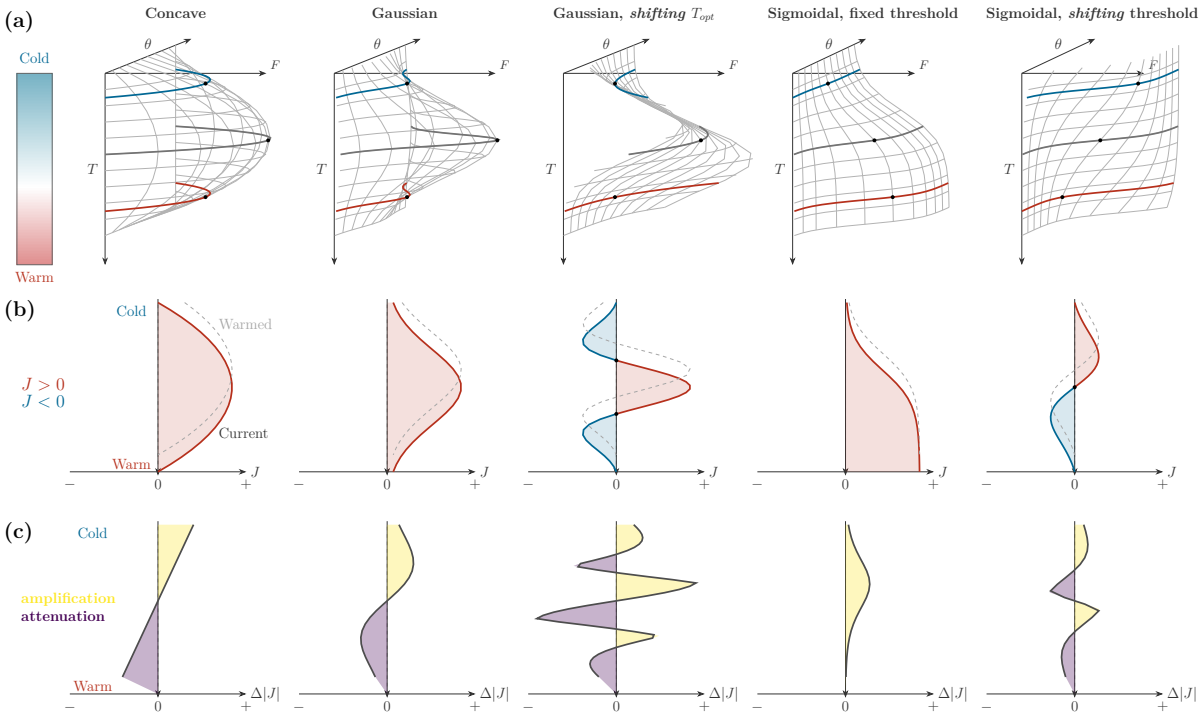


Figure 1: Jensen's inequality and its implications for forecasting biological responses under climate change. For a nonlinear function the value at the mean $f(\bar{\theta})$ differs from the mean of the values $\overline{f(\theta)}$, producing a forecasting bias (hereafter Jensen bias, $J = f(\bar{\theta}) - \overline{f(\theta)}$), where $J \approx -\frac{1}{2} f''(\bar{\theta})\,\mathrm{Var}[\theta]$, and whose sign and magnitude follow the local curvature of f (Denny, 2017; Jensen, 1906). **(a) Representative trait–performance surface illustrating common forms of nonlinear responses to temperature**. The blue-to-red bar on the far left represents an environmental temperature gradient (cold to warm). Performance (F) is shown as a function of trait value (θ) and temperature (T) for five illustrative types: concave dome, Gaussian, Gaussian with shifting thermal optimum (T_{opt}), sigmoidal with fixed threshold, and sigmoidal with shifting threshold. Coloured vertical slices show the trait–performance relationship at cold (*blue*), intermediate (*black*) and warm (*red*) temperatures, while *black points* denote the mean trait value. **(b) Temperature-dependent Jensen bias predicted for each response surface**. *Red shading* indicates positive bias ($J > 0$), where predictions based on the mean trait overestimate performance, whereas the *blue shading* indicates negative bias ($J < 0$), where the mean trait underestimates performance. The *solid curve* represents the bias under the current climate, $J(T)$, and the *dashed curve* the bias after warming, $J(T + \Delta T)$, illustrating how climate change shifts the magnitude and sign of forecasting bias for a fixed trait distribution after warming ($\bar{\theta}$ and $\mathrm{Var}[\theta]$ held constant). **(c) Impact of climate warming on the magnitude of Jensen bias**. The *black curve* shows the change in absolute bias, $|J|$, across the temperature gradient. *Yellow shading* denotes amplification ($|J|$ increases), whereas *purple shading* denotes attenuation ($|J|$ decreases). Together, the examples demonstrate that the same warming scenario can either strengthen or weaken forecasting bias depending solely on the curvature and temperature dependence of the underlying trait–performance relationship.

These ideas have important consequences for ecological forecasting. When the relationship between phenotype, environment and performance is nonlinear, relying on an average phenotype for predictions yields different results than accounting for the phenotypic diversity present in real populations, a consequence of Jensen's inequality (Jensen, 1906; Ruel and Ayres, 1999; **Fig. 1a,b**). We define Jensen bias as $J = f(\bar{\theta}) - \overline{f(\theta)}$, i.e., the difference between performance predicted from the average phenotype and mean performance predicted across individual phenotypes. Importantly, J does not arise from physiological variation alone, but from its interaction with the geometry of the underlying performance surface (Denny, 2017, 2019; Laughlin & Messier, 2015): where the local curvature is convex, predictions based on the average phenotype underestimate the

average performance of heterogeneous populations ($J < 0$), whereas where it is concave, they overestimate it ($J > 0$; Bolnick et al., 2011; Ruel and Ayres, 1999). Different performance-surface geometries, as well as different regions within the same performance surface (e.g., along environmental gradients or across species' geographic ranges), therefore generate distinct patterns of averaging bias (**Fig. 1b**). Because climate change alters environmental conditions (Simon & Amarasekare, 2024) and thus where populations lie on the performance surface, Jensen bias may be amplified, attenuated or even reversed under warming (**Fig. 1c**).

Here, we combined longitudinal measurements of growth and metabolic rates from individually monitored alpine newts (*Ichthyosaura alpestris*; data from Baškiera and Gvoždík, 2021) with coupled Dynamic Energy Budget (DEB) and biophysical models. This framework allowed us to evaluate how among-individual variation in physiological traits influences climate-change forecasts across the species' geographic distribution. DEB theory links environmental conditions to energy acquisition, allocation and expenditure throughout the life cycle (Kooijman, 2010), while biophysical models reconstruct the thermal and hydric environments experienced by organisms from local microclimates (Kearney & Porter, 2009; Kearney et al., 2013). Coupling these approaches therefore allows empirically measured physiological variation to propagate through nonlinear phenotype–environment–performance relationships under current and future climates (**Fig. 1**). By comparing forecasts generated from population-average and individual-level physiological parameterisations under identical environmental simulations, we directly quantified the forecasting bias introduced by physiological averaging. We predicted that (1) forecasts based on an average physiological phenotype would diverge from those incorporating individual-level physiological variation where phenotype–environment–performance relationships were strongly nonlinear; (2) the magnitude and direction of this bias would differ among performance components and across environmental gradients because of differences in local performance-surface geometry; and (3) climate change would amplify, attenuate or reverse these biases by shifting populations into different regions of the performance surface. By linking empirical estimates of individual physiological variation to the geometry of trait–environment–performance surfaces, this study quantifies when and where physiological averaging matters most for ecological forecasts under climate change.

2 Materials and Methods

2.1 Study system and data

We used the alpine newt (*I. alpestris*) as a model species, a temperate amphibian widely distributed across Europe. Juveniles are strictly terrestrial: they usually leave their aquatic larval habitat and disperse on land until sexual maturity (Denoël, 2004). Adults spend most of the year on land except the breeding season (April–June). Terrestrial individuals are predominantly nocturnal or crepuscular and use a variety of daytime retreats, including leaf litter, stones, dead tree trunks and burrows of other animals (Roček et al., 2003). Juveniles, as other predominantly nocturnal amphibians (Enriquez-Urzelai et al., 2020), can show diurnal activity (especially after metamorphosis) to acquire energy reserves before their first winter. The preferred body temperature of the species is 17.3 °C, with a lower bound of thermal preference of 13.8 °C and an upper bound of 20.0 °C (Gvoždík & Kristín, 2017), although individuals remain capable of locomotion at temperatures

approaching 1 °C (Baškiera & Gvoždík, unpublished data).

We compiled two complementary datasets describing the life history and physiological variation of the species. The first dataset consisted of species-level life-history and growth information compiled from the literature: these data included lifespan (20 years), snout–vent length at metamorphosis (1.83 cm), female and male snout–vent length at sexual maturity (4.9 and 4.4 cm, respectively), asymptotic snout–vent length (5.84 cm females, 5.05 cm males), asymptotic body mass (3.16 g females, 2.04 g males), and maximum reproductive rate (600 eggs year^{-1}) (Gvoždík & Kristín, 2017; Miaud et al., 2000). These data were complemented with published growth (time–snout-vent-length) trajectories (Miaud et al., 2000). The second dataset consisted of longitudinal measurements from 37 alpine newts (18 females and 19 males) from a population near Jihlava, Czech Republic (49.2247° N, 16.0579° E; 600 m a.s.l.). These individuals were monitored annually from 2014 (year of metamorphosis) to 2018. Each year, body mass (in g) and standard metabolic rate (in mL O_2 h^{-1}) were measured, providing growth and metabolic trajectories for each individual (Baškiera & Gvoždík, 2021).

2.2 DEB model parameter estimation

Dynamic Energy Budget (DEB) theory (Kooijman, 2010; Sousa et al., 2008) describes the acquisition, storage, and allocation of energy and mass throughout the life cycle. The standard DEB model (Marques et al., 2018) represents an organism by three state variables: structural volume V, reserve energy E, and cumulative energy invested in maturation E_H (plus a reproduction buffer E_R in adults). Energy is assimilated into the reserve at a rate proportional to surface area and mobilised from it to fuel somatic maintenance, growth, maturity maintenance, and maturation (in juveniles) or reproduction (in adults), with life-history transitions occurring at maturity thresholds, e.g., birth at E_H^b and sexual maturity at E_H^p (Kooijman, 2010; Marques et al., 2018; Oliveira et al., 2026). The model is governed by 12 primary parameters (Kooijman, 2010). Among them, the zoom factor z scales maximum body size and the maximum surface-specific assimilation rate $\{\dot{p}_{Am}\}$ (Sousa et al., 2008); reserves are mobilised at a rate set by the energy conductance $\dot{v}$; and, following the κ-rule, a fraction κ of the mobilised reserve is allocated to somatic maintenance, at a cost $[\dot{p}_M]$ per unit of structural volume, and growth, while the remainder supports maturation or reproduction. Because each parameter quantifies the rate or efficiency of a specific physiological process, the model provides a coherent framework for interpreting individual variation in terms of energetic traits with distinct ecological meaning (Kooijman et al., 2021). A full description of the model and its parameters is given in **Appendix 1**.

Because DEB parameters describe latent energetic processes rather than directly measurable biological traits, they were estimated by inverse modelling using the covariation method (Lika et al., 2011; Marques et al., 2018), with the AmPtool and DEBtool_M packages in MATLAB (The MathWorks Inc.). Parameter estimation was performed at two levels: first, a single population-average parameterisation representing the average physiological phenotype (i.e., as is standard practice in mechanistic modelling, a single parameter set fitted to data pooled from multiple individuals; hereafter, average phenotype), and second, separate parameterisations for each of the 37 individuals. For the population-average parameterisation, the model was fitted simultaneously to the species-level life-history information and the complete set of longitudinal growth and metabolic observations from all 37 individuals, which were retained as separate data points

rather than averaged across individuals or ages. We did not average the observations because averaging nonlinear growth and metabolic trajectories produces a composite trajectory that no individual actually follows, which would itself introduce the averaging bias examined here; fitting all observations simultaneously instead yields the parameter set that best describes the pooled population. Of the 12 primary parameters, we estimated seven (z, $\dot{v}$, κ, $[\dot{p}_M]$, the specific cost of structure $[E_G]$, and the maturity thresholds E_H^b and E_H^p), together with a maturity threshold at metamorphosis and male-specific zoom factor (z_m) and puberty threshold ($E_{H,m}^p$) to account for sexual size dimorphism; the remaining parameters were fixed at standard values (Kearney et al., 2013; Kooijman, 2010). Parameter values are given in **Table S2.1**, and further details of the estimation and goodness of fit in **Appendix 1**.

Individual-level parameters (**Table S2.1**) were obtained by re-estimating three parameters independently for each individual: the sex-specific zoom factor (z for females, z_m for males), energy conductance ($\dot{v}$), and volume-specific somatic maintenance ($[\dot{p}_M]$). We allowed these parameters to vary among individuals because they govern the processes captured by our individual-level data: variation in body size and energy uptake (via z and z_m), in energy mobilisation (via $\dot{v}$), and in maintenance costs (via $[\dot{p}_M]$) jointly determine how individuals acquire and use energy, and thus how they grow and respire under a given environment (Kooijman et al., 2021). All remaining parameters, including κ, $[E_G]$, and the maturity thresholds, were held at their population-average values. The estimation used each individual's own longitudinal growth and metabolic trajectories as univariate data, with population-average estimates serving as starting values. All individuals were assigned the same temperature (12 °C) and food availability (scaled functional response $f = 1$) during estimation, so that differences among individual parameter sets reflect differences in growth and metabolism rather than in the conditions experienced. Each individual was assigned the sex-appropriate puberty threshold (E_H^p for females, $E_{H,m}^p$ for males). Consequently, differences among individuals in these parameters can be interpreted as functionally coherent physiological variation.

2.3 Mechanistic modelling framework

To simulate organismal performance under current and future climates, we coupled biophysical and Dynamic Energy Budget (DEB) models using the NicheMapR R package (Kearney & Porter, 2017, 2020). This framework uses biophysical models to compute hourly body temperature (T_b), activity windows, and foraging opportunities given available microclimates and the organism's morphological, physiological and behavioural traits (Kearney et al., 2018; Porter et al., 1973). These hourly values drive the DEB model: T_b is applied to all metabolic rates through the Arrhenius correction, and, since we held food availability constant through the simulations, the time available for foraging determines the amount of energy (food) assimilated, which, together with body temperatures, shapes how organisms grow, develop and reproduce (Kearney, 2012; Kearney & Porter, 2020).

We reconstructed microclimatic conditions with NicheMapR's microclimate model, as implemented in the `micro_global()` function (Kearney & Porter, 2017; Kearney et al., 2014). The model uses monthly climatological means from the global 10 arc-minute climatology of New et al. (2002), together with terrain and soil properties, to estimate hourly air temperature, radiation, wind speed and relative humidity at the animal's height, as well as soil temperature and moisture profiles (Kearney & Porter, 2017; Kearney et al., 2014). Microclimate simulations were run for 22 years with the moisture and snow routines

enabled (Kearney, 2020; Kearney & Maino, 2018), excluding the first two years as spin-up, which resulted in simulations of up to 20 years (i.e., the maximum known lifespan of the species). We ran the microclimate simulations under fully unshaded (0% shade) and deep shade (90% shade) conditions.

To estimate *I. alpestris* performance and simulate its life cycle, we used the `ectotherm()` function of NicheMapR (Kearney & Porter, 2020), parameterised for a terrestrial alpine newt with 90% of the skin surface acting as a free water exchanger (Enriquez-Urzelai & Gvoždík, 2024). Individuals were assumed to be nocturnal and crepuscular, but juveniles were also allowed diurnal activity to acquire energy reserves before their first winter (Enriquez-Urzelai et al., 2020). Animals were active only when body temperature was within their voluntary thermal limits ($VT_{min} = 4\,°C$, $VT_{max} = 20\,°C$) and aboveground relative humidity was at least 65% (Bonnefond et al., 2020; Giacometti & Tattersall, 2025); otherwise, they retreated underground, selecting depths (2.5 cm to 2 m) that kept them within their critical thermal limits ($CT_{min} = -2\,°C$, $CT_{max} = 36\,°C$) (Enriquez-Urzelai & Gvoždík, 2024). When active, individuals could select shaded microhabitats (0–90% shade) to stay close to their preferred body temperature of 17.3 °C (Enriquez-Urzelai et al., 2019; Gvoždík & Kristín, 2017). Females reproduced following the batch reproductive model (Pecquerie et al., 2009), producing clutches of 200–600 eggs, whereas in males the reproductive buffer was continuously depleted during the breeding season (between the spring equinox and the summer solstice). Simulations started at the juvenile stage on 15 August and ran until the individual died (see **Appendix 1** for further details).

Forecasts were generated for every terrestrial cell of a 10 arc-minute grid covering Europe (10° W–40° E, 35°–70° N). For each grid cell, the coupled ectotherm–DEB model was run using the population-average parameterisation (female and male) and each of the 37 individual parameterisations (18 females and 19 males). The population-average and individual-level DEB parameter sets included the appropriate sex-specific zoom factors (z, z_m) and puberty thresholds (E_H^p, $E_{H,m}^p$).

To evaluate the thermal effects of climate change, the complete simulation framework was repeated under two warming scenarios corresponding to uniform increases of +2 °C and +4 °C in hourly air temperature relative to the current climate. These scenarios encompass the range of projected warming across Europe under intermediate to high emission pathways (Calvin et al., 2023). Warming was implemented as a uniform additive offset to the baseline climate's temperature forcing variable, whereas all other climatic variables (precipitation, relative humidity, wind speed, cloud cover and solar radiation) remained unchanged. This approach isolates the direct thermal effects of warming on ectotherm energetics while allowing direct comparison among climate scenarios. Note, however, that because relative humidity was held constant, warmer air implies a higher vapour pressure deficit and, therefore, higher evaporative water loss.

2.4 Quantifying Jensen bias

We quantified Jensen bias as the difference between the prediction obtained from the average phenotype (i.e., the population-average parameterisation) and the mean of the predictions obtained from the individual phenotypes (i.e., the individual-level parameterisations). For every grid cell, physiological parameterisation and climate scenario, we extracted 12 model outputs describing life-history and physiological performance: lifespan (year), lifetime reproductive output (number of eggs in females, hereafter lifetime fecundity; reproductive energy, J, in males), net reproductive rate (R_0), maximum and

minimum annual body temperatures ($T_{b,max}$, $T_{b,min}$), maximum body mass (g), maximum snout–vent length (cm), minimum reserve energy (J), cumulative basking time (h), cumulative foraging time (h), cumulative food intake (g, dry mass) and mean daily water need (g day^{-1}). Cumulative metrics were summed over each individual's entire simulated lifetime. Following NicheMapR terminology, basking time refers to the hours an individual spends above ground without foraging; for a predominantly nocturnal amphibian such as *I. alpestris*, it therefore represents surface exposure for thermoregulation rather than heliothermic basking *sensu stricto*.

For each performance metric, we generated four prediction maps: two population-average predictions (P_{pop}, for females and males) obtained from the population-average parameterisation, and two individual-level predictions obtained by averaging the outputs from the individual parameterisations ($\overline{P}_{\text{ind}}$, for female and male individuals). Forecasting bias associated with physiological averaging was quantified for every grid cell as

$$J = P_{\text{pop}} - \overline{P}_{\text{ind}}$$

where J measures the discrepancy between the population-average prediction and the mean of the individual-level predictions. Positive values of J indicate that the population-average parameterisation overpredicts performance relative to the mean of individuals, whereas negative values indicate underprediction. We computed J for females and males separately. Because performance metrics differ in their units, we compared the magnitude of bias among metrics using the normalised absolute Jensen bias, i.e., the mean $|J|$ across grid cells divided by the mean $|\overline{P}_{\text{ind}}|$ across grid cells ($\times 100$).

To quantify spatial variation in the relative among-individual variability of predicted performance, independently of Jensen bias, we also calculated the coefficient of variation (CV) across the individual-level predictions of each sex (18 females and 19 males), for each metric and grid cell, and compared its median among southern ($<40^\circ$ N), central and northern ($>60^\circ$ N) latitudes. High CV values identify geographical regions and physiological metrics where individual differences translate into the most divergent predicted outcomes.

To evaluate how climate warming modified forecasting bias, we calculated the absolute Jensen bias ($|J|$) for each climate scenario and computed

$$\Delta|J| = |J_{\text{future}}| - |J_{\text{current}}|$$

Positive values indicate amplification of forecasting bias under warming, whereas negative values indicate attenuation. For each metric, we summarised the change in bias as the relative change in absolute Jensen bias (rel. $\Delta|J|$), i.e., the mean $\Delta|J|$ across grid cells divided by the mean $|J_{\text{current}}|$ across grid cells ($\times 100$). We additionally quantified the proportion of grid cells in which $|J|$ increased, decreased or remained unchanged, as well as the proportion in which the sign of J (overprediction versus underprediction) reversed between climate scenarios.

2.5 Consequences of Jensen bias for forecasting

To examine whether Jensen bias varied across environmental gradients, we related the change in absolute Jensen bias ($\Delta|J|$) under $+4\,^\circ$C warming to mean annual air temperature ($\overline{T}_a$), calculated for each grid cell from the climatology of New et al. (2002). For each metric and sex, $\Delta|J|$ was summarised as binned means in $1\,^\circ$C $\overline{T}_a$ intervals, and the

overall trend was described with a generalised additive model (GAM) with a smooth term for $\overline{T}_a$ using the `mgcv` R package (Wood, 2025). Because model outputs are deterministic and neighbouring grid cells are spatially autocorrelated, GAMs were used as descriptive smoothers rather than for statistical inference (model specification in **Appendix 1**).

To evaluate the ecological implications of physiological averaging more directly, we compared forecasts generated from the population-average and individual-level physiological parameterisations for three representative life-history traits, summarising survival, growth and reproduction: lifespan, maximum body mass and lifetime fecundity (or reproductive energy for males). For each trait, grid cell and climate scenario, we compared the population-average prediction with the corresponding mean individual-level prediction, both under the current climate and as projected changes under warming ($P_{\text{future}} - P_{\text{current}}$). We report the percentage of grid cells in which the population-average model under- or overpredicted the individual-level mean, and the percentage in which both parameterisations projected the same direction of change. In addition, we computed the relative bias, $(P_{\text{pop}} - \overline{P}_{\text{ind}})/|\overline{P}_{\text{ind}}| \times 100$, where positive values indicate overestimation by the population-average model. Because relative bias is undefined or unstable where the individual-level mean is zero or close to zero, we summarised it across the species' range by its median and by the 95th percentile of its absolute value (i.e., the magnitude below which 95% of grid-cell estimates fall).

3 Results

3.1 Individual physiological variation

Individual DEB parameters varied across the 37 individuals, with coefficients of variation (CV) spanning from 1.7% (z) to 8% ($[\dot{p}_M]$; **Fig. S2.1**). Among the individually estimated parameters, the volume-specific somatic maintenance cost $[\dot{p}_M]$ was the most variable (CV = 8%), followed by energy conductance $\dot{v}$ (CV = 5.6%). The zoom factors (z: CV = 1.7%; z_m: CV = 1.9%) were less variable. The distribution of each parameter across the 37 individuals appeared approximately normal around the population-average estimate (**Fig. S2.2**). In addition, comparisons between observed and DEB-predicted growth trajectories (**Fig. S2.3**) and metabolic rates (**Fig. S2.4**) confirm that both population-average and individual-level parameterisations reproduced the empirical data accurately. Example 20-year DEB simulations for the population-average parameterisation (female and male) and selected individuals are shown in **Figs. S2.5–S2.14**.

3.2 Physiological averaging generates Jensen bias

Under the current climate, predictions generated from the population-average DEB parameterisation differed systematically from the mean of the individual-level predictions for most performance metrics, although the direction and magnitude of these differences varied among metrics. For females, the population-average parameterisation underestimated most performance metrics relative to the individual-level mean (**Fig. 2**), including basking time ($|J| = 419\,\text{h}$), foraging time ($|J| = 143\,\text{h}$), lifetime fecundity ($|J| = 63\,\text{eggs}$), food intake ($|J| = 3.7\,\text{g}$) and maximum body mass ($|J| = 0.2\,\text{g}$; here and below, reported values are the absolute value of the mean J across grid cells). Relative to the individual-level mean, bias was largest for R_0 (normalised $|J| = 13.3\%$), lifetime fecundity (10.1%) and maximum body mass (5.3%), and smallest for body temperatures ($\leq 0.2\%$). Only

maximum and minimum body temperature showed a consistent positive bias, but with negligible magnitudes (mean $|J| < 0.004\,^{\circ}\mathrm{C}$). The pattern for males was largely opposite: the population-average parameterisation overestimated most metrics, including reproductive energy ($|J| = 2315\,\mathrm{J}$), basking time ($|J| = 286\,\mathrm{h}$) and foraging time ($|J| = 398\,\mathrm{h}$) (**Fig. S3.1**). In relative terms, male bias was largest for reproductive energy (normalised $|J| = 9.1\%$), R_0 (7.4%) and food intake (6.9%).

The direction of Jensen bias differed among ecological performance metrics. Underestimation dominated across most of the geographic range for body size, reserve, food, lifetime fecundity and basking metrics ($J < 0$ in 63–98% of cells), whereas maximum and minimum body temperature showed the opposite pattern, with predominantly positive bias ($J > 0$ in 65–83% of cells). Foraging time showed a more balanced distribution (52% positive, 46% negative and 2% with no bias). For males, the pattern was reversed: overestimation dominated for body size, reserve, food, reproductive energy and basking metrics ($J > 0$ in 57–96% of cells), whereas maximum and minimum body temperature showed predominantly negative bias ($J < 0$ in 84–93% of cells).

The magnitude of physiological heterogeneity also varied geographically (**Figs. S3.2, S3.3**). Compared with central latitudes (40°–60° N), the median coefficient of variation among individual-level predictions was higher towards the southern (<40° N) or northern (>60° N) margins of the modelled area, but the strength of this pattern differed markedly among metrics and between sexes. In females, the median CV at the southern margin was 3.3-fold higher than at central latitudes for lifetime fecundity, 3.1-fold for minimum reserve

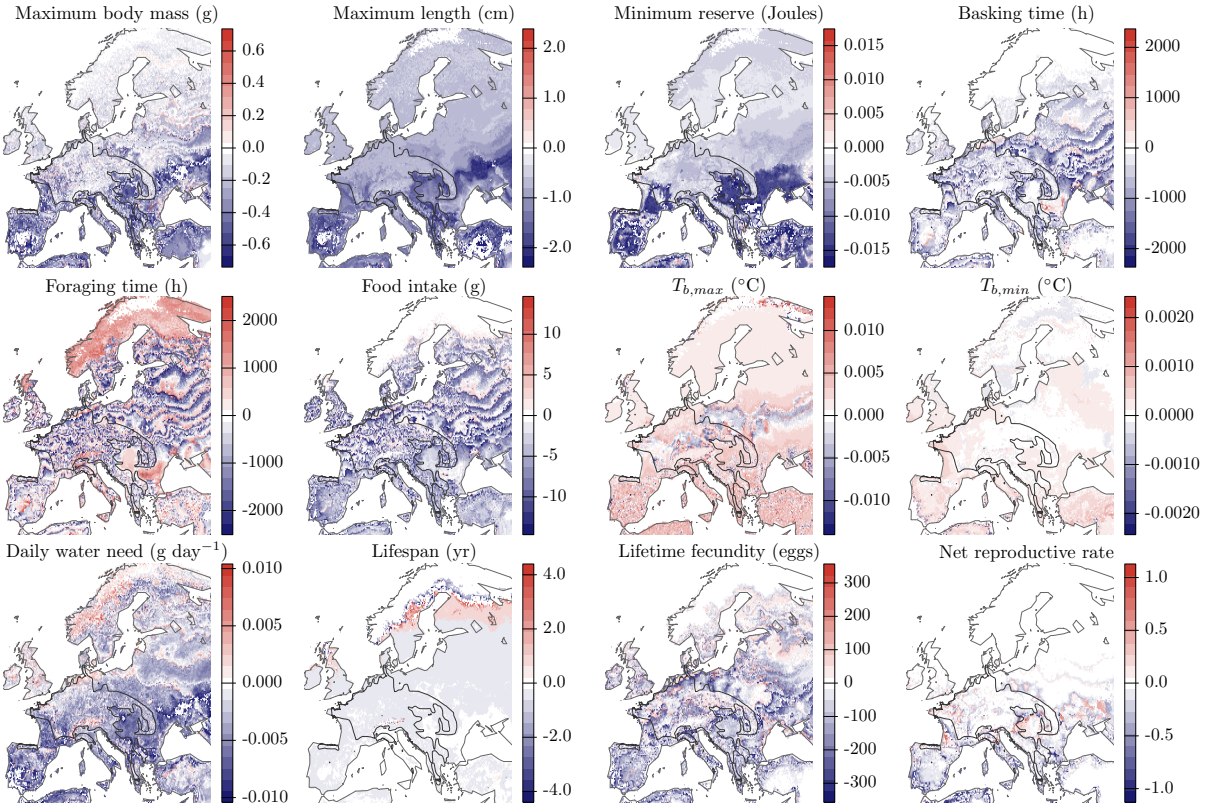


Figure 2: Jensen bias (J) between population-average and mean individual-level prediction surfaces for females under the current climate. Negative values indicate underprediction by the population-average parameterisation, whereas positive values indicate overprediction; values close to zero indicate close agreement. The colour scale is symmetric and centred at zero. The black outline delimits the current distribution of *I. alpestris* (IUCN, 2021). T_b, body temperature.

and 3.4-fold for maximum body length. In males, the largest increases occurred at the northern margin for R_0 (4.3-fold) and reproductive energy (2.2-fold), and at the southern margin for minimum reserve (3.2-fold). In contrast, lifespan, basking time, foraging time and body temperatures differed by at most 1.5-fold between margins and central latitudes in either sex. Thus, the geographic pattern was not universal, but was mainly driven by reproduction- and reserve-related metrics and, in females, by growth-related metrics.

3.3 Climate warming reshapes Jensen bias

Climate warming altered the magnitude of Jensen bias for all performance metrics, but the direction and magnitude of these changes varied among metrics and between sexes (**Figs. 3, S3.4**).

Under +4 °C warming (**Fig. 3**), the absolute Jensen bias $|J|$ changed for all 12 metrics: the strongest relative amplification occurred for minimum body temperature (rel. $\Delta|J| = +568\%$; 72% of cells) and maximum body temperature (rel. $\Delta|J| = +182\%$; 73% of cells), followed by R_0 (rel. $\Delta|J| = +138\%$); in contrast, lifespan (-15%) and minimum reserve (-6%) were attenuated. Note, however, that the large relative changes in body temperature bias arise from very small absolute biases: mean $|J|$ increased from 0.0021 to 0.014 °C for minimum and from 0.0035 to 0.0099 °C for maximum body temperature. For males, the strongest amplification occurred for minimum body temperature (rel. $\Delta|J| = +510\%$; 60% of cells) and maximum body temperature (rel. $\Delta|J| = +215\%$; 62% of cells), whereas lifespan (-1%) was attenuated.

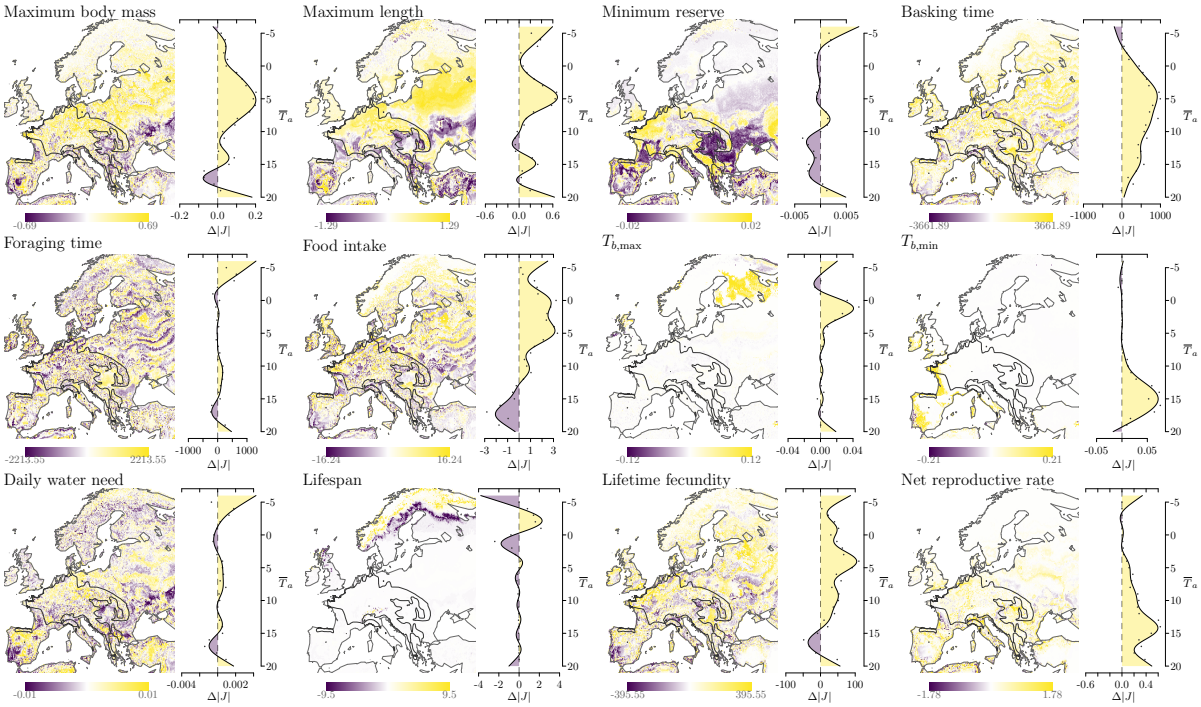


Figure 3: Jensen bias amplification under +4 °C warming. Each map displays the change in absolute Jensen bias ($\Delta|J|$, left, with colour bar) for each performance metric. The curve (descriptive GAM smoother) at the right of each map represents how $\Delta|J|$ changes along the mean annual temperature ($\overline{T}_a$) gradient. Positive values (*yellow*) indicate that warming increases the disagreement between population-average and mean individual-level predictions (amplification), whereas negative values (*purple*) indicate reduced disagreement (attenuation). The black outline delimits the current distribution of *I. alpestris* (IUCN, 2021). T_b, body temperature.

Warming also reversed the direction of Jensen bias in some regions, causing population-average projections to switch between overestimating and underestimating mean individual performance. The sign of J changed in 2–49% of cells between current and +4 °C scenarios, with highest rates for minimum body temperature (49%), foraging time (47%) and R_0 (44%). The spatial maps revealed geographically structured changes in Jensen bias, which were reflected in systematic relationships between $\Delta|J|$ and mean annual temperature. However, both the shape and direction of these temperature relationships differed among ecological performance metrics (**Fig. 3**). For males, the sign of J changed in 5–46% of cells, with highest rates for basking time (46%), foraging time (43%) and minimum reserve (31%).

Even under +2 °C warming, the direction of change was already discernible for most metrics. 11 metrics showed increased $|J|$, with relative changes ranging from +4% (daily water need) to +152% (minimum body temperature), while 1 metric was attenuated: minimum reserve (−5%). 1 metric (lifespan) switched from amplification at +2 °C to attenuation at +4 °C, suggesting a nonlinear threshold above which further warming attenuates rather than amplifies the averaging bias for this trait. For males, 11 metrics showed increased $|J|$, with relative changes ranging from +10% (maximum body length) to +126% (minimum body temperature), while 1 metric was attenuated: minimum reserve (−1%). 1 metric (lifespan) switched from amplification at +2 °C to attenuation at +4 °C.

3.4 Consequences for ecological forecasts

The consequences of physiological averaging for ecological forecasts varied among life-history traits and climate scenarios (**Fig. 4**). The population-average parameterisation systematically underestimated female performance and overestimated male performance for several traits, with differences in both the magnitude and geographical extent of bias. For females, underprediction of lifetime fecundity became more widespread with warming, increasing from 63% of grid cells under the current climate (median relative bias −7.1%) to 71% and 76% of cells under +2 °C and +4 °C, respectively. Similarly, the proportion of cells with underpredicted lifespan increased from 76% under the current climate (median relative bias −1.9%) to 86% under +4 °C. Maximum body mass was underpredicted in 83% of cells (median relative bias −3.8%). Males showed the opposite direction of bias (**Fig. S3.5**): the population-average model overpredicted reproductive energy in 85% of cells under the current climate (median relative bias +8.1%), increasing to 95% of cells under +4 °C (median +10.9%), and also overpredicted lifespan (84% of cells; median +2.1%) and maximum body mass (95% of cells; median +5.5%). Despite these differences, both parameterisations projected the same direction of change under +4 °C warming in at least 90% of grid cells for all three traits in both sexes.

The relative bias maps (**Figs. S3.6, S3.7**) revealed that, for females, the population-average model underestimated lifespan by up to 19% and lifetime fecundity by up to 100% of the individual-level mean, whereas maximum body mass showed relative biases within ±19% for 95% of cells. Under warming, the spatial pattern and magnitude of relative bias shifted, with some regions showing reversed sign. For males, the population-average model overestimated reproductive energy by up to 66% of the individual-level mean, whereas lifespan and maximum body mass showed smaller relative biases.

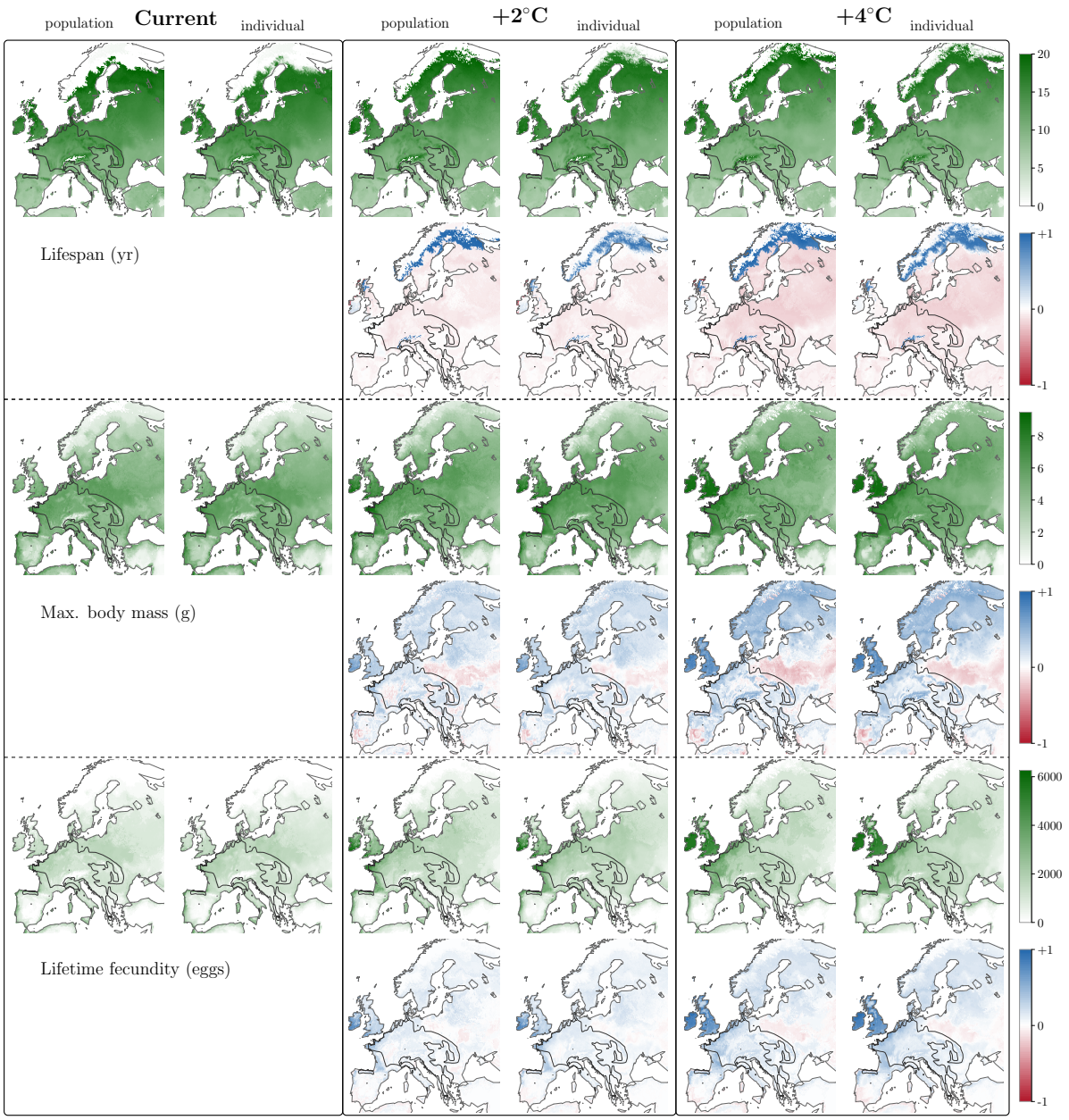


Figure 4: Population-average and mean individual-level predictions and projections of lifespan, maximum body mass and lifetime fecundity under the current climate and +2 °C and +4 °C warming scenarios. For each trait, the top row shows absolute predictions from the population-average parameterisation and the mean of the individual-level parameterisations under each climate scenario, with all six maps displayed on a common trait-specific scale (*green*). The bottom row shows the projected change under +2 °C and +4 °C warming relative to the corresponding current-climate prediction ($P_{\text{future}} - P_{\text{current}}$). For visualisation, projected changes were normalised within each trait by the maximum absolute change across all grid cells, parameterisation levels, and warming scenarios, yielding a common scale from -1 to 1. Negative values (*red*) indicate decreases under warming, positive values (*blue*) indicate increases, and values near zero indicate little change. The black outline delimits the current distribution of *I. alpestris* (IUCN, 2021).

4 Discussion

Mechanistic models have become central to forecasting biological responses to climate change because they explicitly link abiotic environmental conditions to organismal performance through physiology (Fenollosa et al., 2025; Pincebourde & Woods, 2020; Reyes-Puig

et al., 2026). Yet, in most cases species are represented by a single average phenotype. Our results show that this simplification introduces systematic biases that differ among ecological traits, across geographical space and under climate warming. These discrepancies arise because averaging physiological parameters before nonlinear organism–environment interactions does not necessarily yield the same predictions as averaging individual performance afterwards, consistent with Jensen's inequality (Bolnick et al., 2011; Ruel & Ayres, 1999). Extending previous studies on nonlinear responses to environmental variability (Denny, 2017; Paaijmans et al., 2013), we show that variation in only three ecophysiological traits can generate systematic differences in continental-scale mechanistic forecasts. Rather than diminishing the value of mechanistic models, these findings demonstrate how incorporating among-individual physiological variation can help identify where predictions based on average phenotypes are robust and where they are susceptible to bias.

Among-individual physiological variation generated systematic, directional bias whose magnitude and, in some cases, even its sign depended on the ecological trait, sex and environmental context (**Figs. 2, S3.1**) (Bolnick et al., 2011; Denny, 2019; Ruel & Ayres, 1999). In both sexes, reproductive output showed the largest relative bias, whereas the direction of bias was largely opposite between females (predominantly underestimation) and males (predominantly overestimation). Foraging time occupied an intermediate position, repeatedly shifting between over- and underestimation across the species' range, indicating that even the direction of Jensen bias is determined by local environmental conditions rather than by the trait itself (Martin & Huey, 2008). Together, these patterns indicate that traits integrating cumulative energy allocation over the life cycle, particularly reproduction, are considerably more sensitive to physiological averaging than traits dominated by structural growth, such as body mass (Nisbet et al., 2012; Palmer et al., 2024). Reproductive output integrates energy acquisition, allocation and maturation history throughout adulthood (Jusup et al., 2017; Kooijman, 2010), allowing relatively small physiological differences to accumulate through time, whereas body mass is constrained by tighter feedbacks between energy acquisition and maintenance (Lika et al., 2019; Nisbet et al., 2012) that might buffer the consequences of individual variation. The opposite direction of bias observed in females and males may partly arise from their contrasting reproductive allocation strategies, with batch reproduction amplifying energetic differences among females and continuous depletion of the reproductive buffer moderating them in males (Muller et al., 2019; Pecquerie et al., 2009).

Physiological heterogeneity also had geographically structured consequences, indicating that the importance of individual variation depends on where populations occur within environmental space. Variation among the individual-level predictions was generally highest towards the northernmost and southernmost latitudes represented in the simulations, particularly for reproduction- and reserve-related metrics (**Figs. S3.2, S3.3**), showing that comparatively small physiological differences among individuals became increasingly important under environmentally constraining conditions. This result is consistent with theoretical predictions that the ecological consequences of trait variation emerge from the interaction between the amount of variation present and the local geometry of the underlying performance surface, which changes across environmental gradients (Ross et al., 2023). In other words, the same degree of physiological variation can have relatively small ecological consequences in some environments but substantially larger ones in others (Laughlin & Messier, 2015; Li et al., 2021), depending on where populations operate on the performance surface (Polazzo et al., 2024). Although the strongest effects occurred beyond parts of the current realised distribution of *I. alpestris* (IUCN, 2021), they emerged

under the most environmentally challenging conditions represented in the simulations, consistent with the expectation that ecological responses become increasingly sensitive as populations approach environmental limits (Holt, 2003; Sexton et al., 2009; Valladares et al., 2014). This pattern suggests that forecasts based on population-average physiology are therefore most likely to diverge from forecasts that explicitly represent individual variation precisely in situations where predicting persistence, range expansion and local extinction is most relevant (DeMarche et al., 2019).

Furthermore, climate warming altered not only ecological performance itself, but also the magnitude and, in some cases, even the direction of the bias introduced by physiological averaging (**Figs. 3, S3.4**). Climate warming changed Jensen bias for all twelve ecological performance metrics in both sexes, but these changes were strongly trait dependent rather than uniform: in both sexes, warming amplified the bias in net reproductive rate, the most ecologically relevant amplification, whereas it attenuated the bias in lifespan (and, in females, in minimum reserve energy). Body temperatures showed the largest relative amplification, although their biases remained negligible in absolute terms. Warming also reversed the sign of Jensen bias across large parts of the species' range for some metrics, showing that whether population-average parameterisations over- or underestimate ecological performance depends on the environmental conditions under which populations are evaluated. Moreover, whereas most metrics already showed increased bias under +2 °C, lifespan switched from amplification at +2 °C to attenuation at +4 °C, demonstrating that the influence of physiological heterogeneity on ecological forecasts does not change monotonically with warming. Instead, warming can shift populations across different regions of nonlinear performance surfaces (Angilletta Jr., 2009; Kingsolver & Woods, 2016; Li et al., 2021), such that further temperature increases weaken rather than strengthen the effects of physiological averaging. These results are consistent with the broader view that biological responses to climate change emerge from nonlinear interactions between organismal traits and environmental conditions, so that warming changes where populations operate on underlying performance surfaces rather than simply scaling biological responses (Buckley et al., 2018, 2023; Cannistra & Buckley, 2024). Our results show that these same nonlinearities also determine where forecasts based on average physiology provide robust predictions and where they systematically diverge from forecasts that explicitly account for physiological heterogeneity.

Despite the systematic biases introduced by physiological averaging, both physiological parameterisations converged on the same broad biological response of *I. alpestris* to climate warming, projecting the same direction of change in at least 90% of grid cells for lifespan, body mass and reproduction (**Figs. 4, S3.5**). Under both parameterisations, warming reduced projected lifespan across most of the species' distribution, consistent with the faster life histories and higher cumulative metabolic costs expected for ectotherms under warmer conditions (Burraco et al., 2020; Munch & Salinas, 2009). At the same time, both approaches projected reduced body mass and reproductive output towards the north-eastern limit of the species' range, where climatic warming is expected to increase thermal suitability. This combination suggests that improved thermal conditions alone may not translate directly into greater demographic potential if gains in climatic suitability are accompanied by reductions in growth and reproductive performance (Csergő et al., 2017; Schulte, 2015). Thus, incorporating individual physiological variation did not fundamentally alter the biological interpretation of the species' response to climate change, but instead revealed where and by how much forecasts became sensitive to the simplifying assumption of representing populations by a single physiological phenotype. Because mech-

anistic models explicitly represent organismal physiology, they are particularly well suited to incorporate intraspecific physiological variation, phenotypic plasticity and evolutionary responses directly into ecological predictions (Riddell et al., 2023). Previous studies have propagated observed or hypothetical physiological variation through mechanistic models to investigate trait evolution, adaptive significance and thermal adaptation (e.g., De Jong et al., 1996; Enriquez-Urzelai et al., 2020; Kingsolver and Woods, 2016; Riddell et al., 2022); our results show that the same framework can also quantify where the robustness of mechanistic forecasts changes across ecological traits, geographic space and future climates.

As individual-level physiological datasets become increasingly available (Albert et al., 2010; Briscoe et al., 2015; Fyllas et al., 2020; Gardner et al., 2019), propagating trait distributions offers a practical way to quantify the consequences of biological heterogeneity alongside the increasingly realistic representation of environmental heterogeneity. Realising this potential, however, depends on the availability of empirical data describing physiological variation among individuals. Physiological studies commonly report population or treatment means, often accompanied only by summary measures of dispersion, whereas the individual-level estimates on which those summaries are based are rarely reported (Wolf & McKechnie, 2023). Yet, it is precisely the distribution of individual phenotypes, rather than its central tendency, that determines whether and how strongly forecasts based on average phenotypes are biased. Thus, we strongly encourage empiricists to report individual-level estimates (or at least their full distributions and covariances) alongside population means, ideally in open repositories. Such data would enable the incorporation of realistic trait distributions into mechanistic models, allowing us to assess the robustness of ecological forecasts to physiological variation and to identify the environmental conditions and geographical regions where physiological heterogeneity has the greatest consequences for predicted responses to climate change.

Several limitations also point to important directions for future work. We only estimated individual variation for three DEB parameters per individual (the sex-specific zoom factor, $\dot{v}$ and $[\dot{p}_M]$). Variation in the remaining ecophysiological traits or in other traits related to morphology, thermal biology and behaviour could further modify both the magnitude and the spatial pattern of Jensen bias. Likewise, all individuals originated from a single population, whereas physiological trait distributions are likely to vary across species' ranges due to local adaptation, developmental history and long-term acclimation (Enriquez-Urzelai et al., 2020; Moran et al., 2016; Sexton et al., 2009). Incorporating such among-population variation could amplify, attenuate or even reverse the spatial patterns of Jensen bias reported here. Finally, we assumed that physiological differences remained fixed under future climates, whereas phenotypic plasticity and evolutionary responses are expected to reshape trait distributions over the timescales relevant to climate change (Chevin et al., 2010; Kingsolver & Buckley, 2017; Riddell et al., 2023). Extending mechanistic forecasts to account simultaneously for individual variation, geographic differentiation and the dynamic reshaping of trait distributions through plasticity and adaptation therefore represents an important next step towards biologically realistic predictions under environmental change.

5 Conclusion

Our study demonstrates that representing populations by a single average physiological phenotype can introduce systematic, spatially structured biases that depend on the performance metric, sex and environmental conditions, and that are further modified by climate warming. Nevertheless, population-average and individual-level parameterisations generally agreed on the direction of the projected climate response, indicating that physiological averaging can affect quantitative predictions without necessarily changing their qualitative interpretation. These findings highlight an important distinction between the robustness of predicted trends and the accuracy of predicted performance. Incorporating empirically measured physiological variation into mechanistic models therefore offers a means of identifying when average-phenotype assumptions are adequate and when they may obscure ecologically relevant differences. More broadly, accounting for biological heterogeneity alongside environmental heterogeneity can improve our understanding of the conditions under which mechanistic climate forecasts are sensitive to physiological averaging.

Author Contributions

Urtzi Enriquez-Urzelai: writing – original draft, writing – review and editing, conceptualization, formal analysis, methodology, visualization, software. **Gioele Pascucci:** writing – review and editing. **Lumír Gvoždík:** writing – review and editing, conceptualization, funding acquisition.

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Conflicts of Interest

The authors declare no conflicts of interest.

Data Availability Statement

All the data and code to reproduce this study are freely available at GitHub (https://github.com/urtzienriquez/individuals_DEB).

6 References

Albert, C. H., Thuiller, W., Yoccoz, N. G., Soudant, A., Boucher, F., Saccone, P., & Lavorel, S. (2010). Intraspecific functional variability: Extent, structure and sources of variation. *Journal of Ecology*, *98*(3), 604–613. https://doi.org/10.1111/j.1365-2745.2010.01651.x

Angilletta Jr., M. J. (2009, January 29). *Thermal adaptation*. Oxford University Press. https://doi.org/10.1093/acprof:oso/9780198570875.001.1

Baškiera, S., & Gvoždík, L. (2021). Repeatability and heritability of resting metabolic rate in a long-lived amphibian. *Comparative Biochemistry and Physiology Part A: Molecular & Integrative Physiology*, *253*, 110858. https://doi.org/10.1016/j.cbpa.2020.110858

Bolnick, D. I., Amarasekare, P., Araújo, M. S., Bürger, R., Levine, J. M., Novak, M., Rudolf, V. H., Schreiber, S. J., Urban, M. C., & Vasseur, D. A. (2011). Why intraspecific trait variation matters in community ecology. *Trends in Ecology & Evolution*, *26*(4), 183–192. https://doi.org/10.1016/j.tree.2011.01.009

Bolnick, D. I., Svanbäck, R., Fordyce, J. A., Yang, L. H., Davis, J. M., Hulsey, C. D., & Forister, M. L. (2003). The Ecology of Individuals: Incidence and Implications of Individual Specialization. *The American Naturalist*, *161*(1), 1–28. https://doi.org/10.1086/343878

Bonnefond, A., Courtois, E. A., Sueur, J., Sugai, L. S. M., & Llusia, D. (2020). Climatic breadth of calling behaviour in two widespread Neotropical frogs: Insights from humidity extremes. *Global Change Biology*, *26*(10), 5431–5446. https://doi.org/10.1111/gcb.15266

Briscoe, N. J., Krockenberger, A., Handasyde, K. A., & Kearney, M. R. (2015). Bergmann meets Scholander: Geographical variation in body size and insulation in the koala is related to climate (M. McGeoch, Ed.). *Journal of Biogeography*, *42*(4), 791–802. https://doi.org/10.1111/jbi.12445

Briscoe, N. J., Morris, S. D., Mathewson, P. D., Buckley, L. B., Jusup, M., Levy, O., Maclean, I. M. D., Pincebourde, S., Riddell, E. A., Roberts, J. A., Schouten, R., Sears, M. W., & Kearney, M. R. (2023). Mechanistic forecasts of species responses to climate change: The promise of biophysical ecology. *Global Change Biology*, *29*(6), 1451–1470. https://doi.org/10.1111/gcb.16557

Buckley, L. B., Cannistra, A. F., & John, A. (2018). Leveraging organismal biology to forecast the effects of climate change. *Integrative and Comparative Biology*, *58*(1), 38–51. https://doi.org/10.1093/icb/icy018

Buckley, L. B., Carrington, E., Dillon, M. E., García-Robledo, C., Roberts, S. B., Wegrzyn, J. L., & Urban, M. C. (2023). Characterizing biological responses to climate variability and extremes to improve biodiversity projections (M. R. Moura, Ed.). *PLOS Climate*, *2*(6), e0000226. https://doi.org/10.1371/journal.pclm.0000226

Burraco, P., Orizaola, G., Monaghan, P., & Metcalfe, N. B. (2020). Climate change and ageing in ectotherms. *Global Change Biology*, *26*(10), 5371–5381. https://doi.org/10.1111/gcb.15305

Calvin, K., Dasgupta, D., Krinner, G., Mukherji, A., Thorne, P. W., Trisos, C., Romero, J., Aldunce, P., Barrett, K., Blanco, G., Cheung, W. W., Connors, S., Denton, F., Diongue-Niang, A., Dodman, D., Garschagen, M., Geden, O., Hayward, B., Jones, C., … Ha, M. (2023, July 25). *IPCC, 2023: Climate change 2023: Synthesis report. Contribution of working groups I, II and III to the sixth assessment report of the intergovernmental panel on climate change [core writing team, H. Lee and J. Romero (eds.)]. IPCC, geneva, switzerland* (P. Arias, M. Bustamante, I. Elgizouli, G. Flato, M. Howden, C. Méndez-Vallejo, J. J. Pereira, R. Pichs-Madruga, S. K. Rose, Y. Saheb, R. Sánchez Rodríguez, D. Ürge-Vorsatz, C. Xiao, N. Yassaa, J. Romero, J. Kim, E. F. Haites, Y. Jung, R. Stavins, … C. Péan, Eds.; First). Intergovernmental Panel on Climate Change (IPCC). https://doi.org/10.59327/IPCC/AR6-9789291691647

Cannistra, A. F., & Buckley, L. B. (2024). Accounting for nonlinear responses to traits improves range shift predictions. *Ecosphere*, *15*(6), e4860. https://doi.org/10.1002/ecs2.4860

Chevin, L.-M., Lande, R., & Mace, G. M. (2010). Adaptation, plasticity, and extinction in a changing environment: Towards a predictive theory (J. G. Kingsolver, Ed.). *PLOS Biology*, *8*(4), e1000357. https://doi.org/10.1371/journal.pbio.1000357

Csergő, A. M., Salguero-Gómez, R., Broennimann, O., Coutts, S. R., Guisan, A., Angert, A. L., Welk, E., Stott, I., Enquist, B. J., McGill, B., Svenning, J.-C., Violle, C., & Buckley, Y. M. (2017). Less favourable climates constrain demographic strategies in plants (J. Gurevitch, Ed.). *Ecology Letters*, *20*(8), 969–980. https://doi.org/10.1111/ele.12794

De Jong, P. W., Gussekloo, S. W. S., & Brakefield, P. M. (1996). Differences in Thermal Balance, Body Temperature and Activity Between Non-Melanic and Melanic Two-Spot Ladybird Beetles (*Adalia Bipunctata*) Under Controlled Conditions. *Journal of Experimental Biology*, *199*(12), 2655–2666. https://doi.org/10.1242/jeb.199.12.2655

DeMarche, M. L., Doak, D. F., & Morris, W. F. (2019). Incorporating local adaptation into forecasts of species' distribution and abundance under climate change. *Global Change Biology*, *25*(3), 775–793. https://doi.org/10.1111/gcb.14562

Denny, M. (2017). The fallacy of the average: On the ubiquity, utility and continuing novelty of Jensen's inequality. *Journal of Experimental Biology*, *220*(2), 139–146. https://doi.org/10.1242/jeb.140368

Denny, M. (2019). Performance in a variable world: Using Jensen's inequality to scale up from individuals to populations (P. Domenici, Ed.). *Conservation Physiology*, *7*(1), coz053. https://doi.org/10.1093/conphys/coz053

Denoël, M. (2004). Terrestrial*versus*aquatic foraging in juvenile Alpine newts (*Triturus alpestris*). *Écoscience*, *11*(4), 404–409. https://doi.org/10.1080/11956860.2004.11682849

Des Roches, S., Post, D. M., Turley, N. E., Bailey, J. K., Hendry, A. P., Kinnison, M. T., Schweitzer, J. A., & Palkovacs, E. P. (2018). The ecological importance of intraspecific variation. *Nature Ecology & Evolution*, *2*(1), 57–64. https://doi.org/10.1038/s41559-017-0402-5

Elith, J., & Leathwick, J. R. (2009). Species distribution models: Ecological explanation and prediction across space and time. *Annual Review of Ecology, Evolution, and Systematics*, *40*(1), 677–697. https://doi.org/10.1146/annurev.ecolsys.110308.120159

Enriquez-Urzelai, U., & Gvoždík, L. (2024). Impacts of behaviour and acclimation of metabolic rate on energetics in sheltered ectotherms: A climate change perspective. *Proceedings of the Royal Society B: Biological Sciences*, *291*(2017), 20232152. https://doi.org/10.1098/rspb.2023.2152

Enriquez-Urzelai, U., Kearney, M. R., Nicieza, A. G., & Tingley, R. (2019). Integrating mechanistic and correlative niche models to unravel range-limiting processes in a temperate amphibian. *Global Change Biology*, *25*(8), 2633–2647. https://doi.org/10.1111/gcb.14673

Enriquez-Urzelai, U., Tingley, R., Kearney, M. R., Sacco, M., Palacio, A. S., Tejedo, M., & Nicieza, A. G. (2020). The roles of acclimation and behaviour in buffering climate change impacts along elevational gradients (K. Marske, Ed.). *Journal of Animal Ecology*, *89*(7), 1722–1734. https://doi.org/10.1111/1365-2656.13222

Fenollosa, E., Pang, S., Briscoe, N., Guisan, A., & Salguero-Gomez, R. (2025). Powerful yet challenging: Mechanistic Niche Models for predicting invasive species potential distribution under climate change. *Ecography*. https://doi.org/10.32942/X26341

Fyllas, N. M., Michelaki, C., Galanidis, A., Evangelou, E., Zaragoza-Castells, J., Dimitrakopoulos, P. G., Tsadilas, C., Arianoutsou, M., & Lloyd, J. (2020). Functional Trait Variation Among and Within Species and Plant Functional Types in Mountainous Mediterranean Forests. *Frontiers in Plant Science*, *11*, 212. https://doi.org/10.3389/fpls.2020.00212

Gardner, J. L., Amano, T., Peters, A., Sutherland, W. J., Mackey, B., Joseph, L., Stein, J., Ikin, K., Little, R., Smith, J., & Symonds, M. R. E. (2019). Australian songbird body size tracks climate variation: 82 species over 50 years. *Proceedings of the Royal Society B: Biological Sciences*, *286*(1916), 20192258. https://doi.org/10.1098/rspb.2019.2258

Giacometti, D., & Tattersall, G. J. (2025). Behavioural evidence of a humidistat: A temperature-compensating mechanism of hydroregulation in spotted salamanders. *Journal of Experimental Biology*, *228*(10), jeb250297. https://doi.org/10.1242/jeb.250297

Gvoždík, L. (2024). Individual variation in thermally induced plasticity of metabolic rates: Ecological and evolutionary implications for a warming world. *Philosophical Transactions of the Royal Society B: Biological Sciences*, *379*(1896), 20220494. https://doi.org/10.1098/rstb.2022.0494

Gvoždík, L., & Kristín, P. (2017). Economic thermoregulatory response explains mismatch between thermal physiology and behavior in newts. *Journal of Experimental Biology*, *220*(6), jeb.145573. https://doi.org/10.1242/jeb.145573

Holt, R. D. (2003). On the evolutionary ecology of species' ranges. *Evolutionary Ecology Research*, *5*, 159–178.

IUCN. (2021, July 9). *Ichthyosaura alpestris: IUCN SSC Amphibian Specialist Group: The IUCN Red List of Threatened Species 2022: E.T59472A89702213*. International Union for Conservation of Nature. https://doi.org/10.2305/IUCN.UK.2022-1.RLTS.T59472A89702213.en

Jensen, J. L. W. V. (1906). Sur les fonctions convexes et les inégalités entre les valeurs moyennes. *Acta Mathematica*, *30*(0), 175–193. https://doi.org/10.1007/bf02418571

Jusup, M., Sousa, T., Domingos, T., Labinac, V., Marn, N., Wang, Z., & Klanjšček, T. (2017). Physics of metabolic organization. *Physics of Life Reviews*, *20*, 1–39. https://doi.org/10.1016/j.plrev.2016.09.001

Kearney, M. (2012). Metabolic theory, life history and the distribution of a terrestrial ectotherm. *Functional Ecology*, *26*(1), 167–179. https://doi.org/10.1111/j.1365-2435.2011.01917.x

Kearney, M., & Porter, W. (2009). Mechanistic niche modelling: Combining physiological and spatial data to predict species' ranges. *Ecology Letters*, *12*(4), 334–350. https://doi.org/10.1111/j.1461-0248.2008.01277.x

Kearney, M. R. (2020). How will snow alter exposure of organisms to cold stress under climate warming? (A. Guisan, Ed.). *Global Ecology and Biogeography*, *29*(7), 1246–1256. https://doi.org/10.1111/geb.13100

Kearney, M. R., & Maino, J. L. (2018). Can next-generation soil data products improve soil moisture modelling at the continental scale? An assessment using a new microclimate package for the R programming environment. *Journal of Hydrology*, *561*, 662–673. https://doi.org/10.1016/j.jhydrol.2018.04.040

Kearney, M. R., Munns, S. L., Moore, D., Malishev, M., & Bull, C. M. (2018). Field tests of a general ectotherm niche model show how water can limit lizard activity and distribution. *Ecological Monographs*, *88*(4), 672–693. https://doi.org/10.1002/ecm.1326

Kearney, M. R., & Porter, W. P. (2017). NicheMapR – an R package for biophysical modelling: The microclimate model. *Ecography*, *40*(5), 664–674. https://doi.org/10.1111/ecog.02360

Kearney, M. R., & Porter, W. P. (2020). NicheMapR – an R package for biophysical modelling: The ectotherm and dynamic energy budget models. *Ecography*, *43*(1), 85–96. https://doi.org/10.1111/ecog.04680

Kearney, M. R., Shamakhy, A., Tingley, R., Karoly, D. J., Hoffmann, A. A., Briggs, P. R., & Porter, W. P. (2014). Microclimate modelling at macro scales: A test of a general microclimate model integrated with gridded continental-scale soil and weather data (J. Travis, Ed.). *Methods in Ecology and Evolution*, *5*(3), 273–286. https://doi.org/10.1111/2041-210X.12148

Kearney, M. R., Simpson, S. J., Raubenheimer, D., & Kooijman, S. A. L. M. (2013). Balancing heat, water and nutrients under environmental change: A thermodynamic niche framework (S. Chown, Ed.). *Functional Ecology*, *27*(4), 950–966. https://doi.org/10.1111/1365-2435.12020

Kingsolver, J. G., & Buckley, L. B. (2017). Evolution of plasticity and adaptive responses to climate change along climate gradients. *Proceedings of the Royal Society B: Biological Sciences*, *284*(1860), 20170386. https://doi.org/10.1098/rspb.2017.0386

Kingsolver, J. G., & Woods, H. A. (2016). Beyond Thermal Performance Curves: Modeling Time-Dependent Effects of Thermal Stress on Ectotherm Growth Rates. *The American Naturalist*, *187*(3), 283–294. https://doi.org/10.1086/684786

Kolbe, J. J., Kearney, M., & Shine, R. (2010). Modeling the consequences of thermal trait variation for the cane toad invasion of Australia. *Ecological Applications*, *20*(8), 2273–2285. https://doi.org/10.1890/09-1973.1

Kontopoulos, D. .-. G., Sentis, A., Daufresne, M., Glazman, N., Dell, A. I., & Pawar, S. (2024). No universal mathematical model for thermal performance curves across traits and taxonomic groups. *Nature Communications*, *15*(1), 8855. https://doi.org/10.1038/s41467-024-53046-2

Kooijman, S. A. L. M., Lika, K., Augustine, S., & Marn, N. (2021). Multidimensional scaling for animal traits in the context of dynamic energy budget theory. *Conservation Physiology*, *9*(1), coab086. https://doi.org/10.1093/conphys/coab086

Kooijman, S. A. L. M. (2010). *Dynamic energy budget theory for metabolic organisation* (3rd ed.). Cambridge university press.

Laughlin, D. C., & Messier, J. (2015). Fitness of multidimensional phenotypes in dynamic adaptive landscapes. *Trends in Ecology & Evolution*, *30*(8), 487–496. https://doi.org/10.1016/j.tree.2015.06.003

Li, Y., Jiang, Y., Shipley, B., Li, B., Luo, W., Chen, Y., Zhao, K., He, D., Rodríguez-Hernández, D. I., & Chu, C. (2021). The complexity of trait–environment performance landscapes in a local subtropical forest. *New Phytologist*, *229*(3), 1388–1397. https://doi.org/10.1111/nph.16955

Lika, K., Augustine, S., & Kooijman, S. A. (2019). Body size as emergent property of metabolism. *Journal of Sea Research*, *143*, 8–17. https://doi.org/10.1016/j.seares.2018.04.005

Lika, K., Kearney, M. R., & Kooijman, S. A. (2011). The "covariation method" for estimating the parameters of the standard dynamic energy budget model II: Properties and preliminary patterns. *Journal of Sea Research*, *66*(4), 278–288. https://doi.org/10.1016/j.seares.2011.09.004

Maino, J. L., Kong, J. D., Hoffmann, A. A., Barton, M. G., & Kearney, M. R. (2016). Mechanistic models for predicting insect responses to climate change. *Current Opinion in Insect Science*, *17*, 81–86. https://doi.org/10.1016/j.cois.2016.07.006

Marques, G. M., Augustine, S., Lika, K., Pecquerie, L., Domingos, T., & Kooijman, S. A. L. M. (2018). The AmP project: Comparing species on the basis of dynamic energy budget parameters (D. A. Beard, Ed.). *PLOS Computational Biology*, *14*(5), e1006100. https://doi.org/10.1371/journal.pcbi.1006100

Martin, T. L., & Huey, R. B. (2008). Why "suboptimal" is optimal: Jensen's inequality and ectotherm thermal preferences. *The American Naturalist*, *171*(3), E102–E118. https://doi.org/10.1086/527502

Miaud, C., Guyetant, R., & Faber, H. (2000). Age, Size, and Growth of the Alpine Newt, *Triturus alpestris* (Urodela: Salamandridae), at High Altitude and a Review of Life-History Trait Variation throughout Its Range. *Herpetologica*, *56*(2), 135–144. Retrieved July 15, 2026, from https://www.jstor.org/stable/3893265

Moran, E. V., Hartig, F., & Bell, D. M. (2016). Intraspecific trait variation across scales: Implications for understanding global change responses. *Global Change Biology*, *22*(1), 137–150. https://doi.org/10.1111/gcb.13000

Muller, E. B., Lika, K., Nisbet, R. M., Schultz, I. R., Casas, J., Gergs, A., Murphy, C. A., Nacci, D., & Watanabe, K. H. (2019). Regulation of reproductive processes with dynamic energy budgets (I. Smallegange, Ed.). *Functional Ecology*, *33*(5), 819–832. https://doi.org/10.1111/1365-2435.13298

Munch, S. B., & Salinas, S. (2009). Latitudinal variation in lifespan within species is explained by the metabolic theory of ecology. *Proceedings of the National Academy of Sciences*, *106*(33), 13860–13864. https://doi.org/10.1073/pnas.0900300106

New, M., Lister, D., Hulme, M., & Makin, I. (2002). A high-resolution data set of surface climate over global land areas. *Climate Research*, *21*(1), 1–25. https://doi.org/10.3354/cr021001

Nisbet, R. M., Jusup, M., Klanjscek, T., & Pecquerie, L. (2012). Integrating dynamic energy budget (DEB) theory with traditional bioenergetic models. *Journal of Experimental Biology*, *215*(7), 1246–1246. https://doi.org/10.1242/jeb.071845

Oliveira, D. F., Sulc, A., Moan, E. L., Donati, E., Klagkou, E., Lagunes, M.-J., Dajčman, U., & Tan, T. Y. (2026). How to build Dynamic Energy Budget models: Practical

guidelines for model development. *Ecological Modelling*, *521*, 111703. https://doi.org/10.1016/j.ecolmodel.2026.111703

Paaijmans, K. P., Heinig, R. L., Seliga, R. A., Blanford, J. I., Blanford, S., Murdock, C. C., & Thomas, M. B. (2013). Temperature variation makes ectotherms more sensitive to climate change. *Global Change Biology*, *19*(8), 2373–2380. https://doi.org/10.1111/gcb.12240

Palmer, M., Moro-Martínez, I., Tomàs-Ferrer, J., Grau, A., López-Belluga, M. D., Herlin, M., Stavrakidis-Zachou, O., & Campos-Candela, A. (2024). Assessing between-individual variability in bioenergetics modelling: Opportunities, challenges, and potential applications. *Ecological Modelling*, *498*, 110848. https://doi.org/10.1016/j.ecolmodel.2024.110848

Pecquerie, L., Petitgas, P., & Kooijman, S. A. (2009). Modeling fish growth and reproduction in the context of the dynamic energy budget theory to predict environmental impact on anchovy spawning duration. *Journal of Sea Research*, *62*(2–3), 93–105. https://doi.org/10.1016/j.seares.2009.06.002

Pettersen, A. K., & Metcalfe, N. B. (2024). Consequences of the cost of living: Is variation in metabolic rate evolutionarily significant? *Philosophical Transactions of the Royal Society B: Biological Sciences*, *379*(1896), 20220498. https://doi.org/10.1098/rstb.2022.0498

Pincebourde, S., & Woods, H. A. (2020). There is plenty of room at the bottom: Microclimates drive insect vulnerability to climate change. *Current Opinion in Insect Science*, *41*, 63–70. https://doi.org/10.1016/j.cois.2020.07.001

Polazzo, F., Limberger, R., Pennekamp, F., Ross, S. R. P.-J., Simpson, G. L., & Petchey, O. L. (2024). Measuring the Response Diversity of Ecological Communities Experiencing Multifarious Environmental Change. *Global Change Biology*, *30*(11), e17594. https://doi.org/10.1111/gcb.17594

Porter, W. P., Mitchell, J. W., Beckman, W. A., & DeWitt, C. B. (1973). Behavioral implications of mechanistic ecology: Thermal and behavioral modeling of desert ectotherms and their microenvironment. *Oecologia*, *13*(1), 1–54. https://doi.org/10.1007/BF00379617

Reyes-Puig, C., Kaliontzopoulou, A., Sillero, N., & Enriquez-Urzelai, U. (2026). Physiology–microhabitat matching may help organisms cope with the thermal and hydric challenges under climate change: A tale of two lizards. *Ecography*, *2026*(3), e08337. https://doi.org/10.1002/ecog.08337

Riddell, E. A., Burger, I. J., Tyner-Swanson, T. L., Biggerstaff, J., Muñoz, M. M., Levy, O., & Porter, C. K. (2023). Parameterizing mechanistic niche models in biophysical ecology: A review of empirical approaches. *Journal of Experimental Biology*, *226*(22), jeb245543. https://doi.org/10.1242/jeb.245543

Riddell, E. A., Patton, J. L., & Beissinger, S. R. (2022). Thermal adaptation of pelage in desert rodents balances cooling and insulation. *Evolution*, evo.14643. https://doi.org/10.1111/evo.14643

Roček, Z., Joly, P., & Grossenbacher, K. (2003). Triturus alpestris (Laurenti, 1768) – Bergmolch. In *Handbuch der Reptilien und Amphibien Europas. Band 4/IIA. Schwanzlurche (Urodela) IIA. Salamandridae II: Triturus 1* (pp. 607–656). Wiesbaden, AULA.

Ross, S. R. P.-J., Petchey, O. L., Sasaki, T., & Armitage, D. W. (2023). How to measure response diversity. *Methods in Ecology and Evolution*, *14*(5), 1150–1167. https://doi.org/10.1111/2041-210X.14087

Ruel, J. J., & Ayres, M. P. (1999). Jensen's inequality predicts effects of environmental variation. *Trends in Ecology & Evolution*, *14*(9), 361–366. https://doi.org/10.1016/S0169-5347(99)01664-X

Schulte, P. M. (2015). The effects of temperature on aerobic metabolism: Towards a mechanistic understanding of the responses of ectotherms to a changing environment (J. E. Podrabsky, J. H. Stillman, & L. Tomanek, Eds.). *Journal of Experimental Biology*, *218*(12), 1856–1866. https://doi.org/10.1242/jeb.118851

Seebacher, F., & Franklin, C. E. (2012). Determining environmental causes of biological effects: The need for a mechanistic physiological dimension in conservation biology. *Philosophical Transactions of the Royal Society B: Biological Sciences*, *367*(1596), 1607–1614. https://doi.org/10.1098/rstb.2012.0036

Sexton, J. P., McIntyre, P. J., Angert, A. L., & Rice, K. J. (2009). Evolution and Ecology of Species Range Limits. *Annual Review of Ecology, Evolution, and Systematics*, *40*(1), 415–436. https://doi.org/10.1146/annurev.ecolsys.110308.120317

Simon, M. W., & Amarasekare, P. (2024). Predicting the fundamental thermal niche of ectotherms. *Ecology*, *105*(5), e4289. https://doi.org/10.1002/ecy.4289

Sinclair, B. J., Marshall, K. E., Sewell, M. A., Levesque, D. L., Willett, C. S., Slotsbo, S., Dong, Y., Harley, C. D. G., Marshall, D. J., Helmuth, B. S., & Huey, R. B. (2016). Can we predict ectotherm responses to climate change using thermal performance curves and body temperatures? (D. Vasseur, Ed.). *Ecology Letters*, *19*(11), 1372–1385. https://doi.org/10.1111/ele.12686

Sousa, T., Domingos, T., & Kooijman, S. (2008). From empirical patterns to theory: A formal metabolic theory of life. *Philosophical Transactions of the Royal Society B: Biological Sciences*, *363*(1502), 2453–2464. https://doi.org/10.1098/rstb.2007.2230

Urban, M. C., Bocedi, G., Hendry, A. P., Mihoub, J.-B., Pe'er, G., Singer, A., Bridle, J. R., Crozier, L. G., De Meester, L., Godsoe, W., Gonzalez, A., Hellmann, J. J., Holt, R. D., Huth, A., Johst, K., Krug, C. B., Leadley, P. W., Palmer, S. C. F., Pantel, J. H., … Travis, J. M. J. (2016). Improving the forecast for biodiversity under climate change. *Science*, *353*(6304), aad8466. https://doi.org/10.1126/science.aad8466

Valladares, F., Matesanz, S., Guilhaumon, F., Araújo, M. B., Balaguer, L., Benito-Garzón, M., Cornwell, W., Gianoli, E., Van Kleunen, M., Naya, D. E., Nicotra, A. B., Poorter, H., & Zavala, M. A. (2014). The effects of phenotypic plasticity and local adaptation on forecasts of species range shifts under climate change (W. Thuiller, Ed.). *Ecology Letters*, *17*(11), 1351–1364. https://doi.org/10.1111/ele.12348

Wolf, B. O., & McKechnie, A. E. (2023). Biophysical approaches to predicting species vulnerability. *Global Change Biology*, *29*(6), 1421–1422. https://doi.org/10.1111/gcb.16565

Wood, S. (2025). *Mgcv: Mixed gam computation vehicle with automatic smoothness estimation* [R package version 1.9-4]. https://doi.org/10.32614/CRAN.package.mgcv